

Data Structure Lab(25CS151)

Programs List

1.1	Largest of Three Numbers
Problem Statement	Write an Algorithm to input three numbers and determine which number is the largest among them
Input Format	-
Output Format	-
Sample Input	-
Sample Output	-

1.2

Leap Year

Problem
Statement

Write an Algorithm to check whether the user input year is Leap year or not.

Input
Format

-

Output
Format

-

Sample
Input

-

Sample
Output

-

1.3	Sum of digits
Problem Statement	Draw a Flow-Chart to display the sum of digits of 4 digits user input number.
Input Format	-
Output Format	-
Sample Input	-
Sample Output	-

2.1	Swapping of two numbers
Problem Statement	Write a program to input two numbers and display the numbers after swapping their values.
Input Format	First line of input contains the two integers denoting the value of a and b.
Output Format	Display numbers after swapping in a single line.
Sample Input	5 9
Sample Output	9 5

2.2	Salary Calculation
Problem Statement	Basic salary of an employee is input through the keyboard. The DA is 25% of the basic salary while the HRA is 15% of the basic salary. Provident Fund is deducted at the rate of 10% of the gross salary (BS+DA+HRA). Program to calculate the Net Salary.
Input Format	First line of input contain the integers denoting the value of Basic Salary.
Output Format	Display the Net Salary
Sample Input	10000
Sample Output	12600

2.3	Sum of digits
Problem Statement	Write a program to input 4-digits number and display the sum of digits.
Input Format	First line of input contain the 4 digits integers
Output Format	Display the sum of digits
Sample Input	4587
Sample Output	24

3.1	Largest of Three Numbers	
Problem Statement	Write a program to input three numbers and determine which number is the largest among them.Using Ternary Operator	
Input Format	First line of input contains the three integers denoting the value of a,b and c.	
Output Format	Largest integer in a single line.	
Sample Input	5 9 2	
Sample Output	9	

3.2	Leap Year
Problem Statement	Write a program to input a year and check the input year is leap year or not using ternary Operator.
Input Format	First line contains the integer denotaing the year
Output Format	Display the result either Leap year or Not a Leap Year.
Sample Input	2024
Sample Output	Leap Year

3.3	4 digits Palindrome Number
Problem Statement	Write a program to input a 4 digits number and check the input number is palindrome or not using ternary Operator.
Input Format	First line contains the integer denotaing the 4 digits number
Output Format	Display the result either Yes or No.
Sample Input	1331
Sample Output	Yes

4.1	Calculator
Problem Statement	Write a program to input two numbers and one operation choice(1.Addition, 2.Subtraction, 3. Multiplication, 4. Division).Perform the binary operation as per the given choice.
Input Format	First line contain the integers denotaing the two numbers and operation choice
Output Format	Display the result.
Sample Input	10 20 3
Sample Output	200

4.2	Display Fibonacci Series Until N
Problem Statement	Generate and display the Fibonacci sequence up to a given limit.
Input Format	First line contain the single number.
Output Format	Display the series upto given limit.
Sample Input	50
Sample Output	0 1 1 2 3 5 8 13 21 34

4.3	Check for Armstrong Number
Problem Statement	An Armstrong number of n digits is a number equal to the sum of its digits raised to the power n. Find if a number is an Armstrong number or not.
Input Format	First line contains the value of a number.
Output Format	Display the message Yes or Not.
Sample Input	1634
Sample Output	Yes

5.1	Sum of Digits Until a Single Digit
Problem Statement	Find the sum of digits of a number repeatedly until the sum becomes a single-digit number.
Input Format	First line contain the number.
Output Format	Display the final single digit.
Sample Input	9875
Sample Output	2 (Explanation: $9+8+7+5=29 \rightarrow 2+9=11 \rightarrow 1+1=2$)

5.2	Display All Prime Numbers Between 1 to N
Problem Statement	find and print all prime numbers between 1 and N.
Input Format	First line contain the value of N.
Output Format	Display all space separated prime numbers.
Sample Input	50
Sample Output	2 3 5 7 11 13 17 19 23 29 31 37 41 43 47

5.3	Pattern Display
Problem Statement	Display the given pattern for a given number of rows.
Input Format	First line contain the value of N.
Output Format	Display the pattern.
Sample Input	5
Sample Output	<pre> 1 121 12321 1234321 123454321 </pre>

6.1	Display Top-3 Students
Problem Statement	Display the top three students marks.given that n students marks.
Input Format	First line contain the number of students(n). Second line contain the marks of n students
Output Format	Display the top three student marks.
Sample Input	10 80 87 55 85 82 79 74 95 65 91
Sample Output	95 91 87

6.2	Binary-Search
Problem Statement	Find the location of a given number N in an array whose size and elements are given. The elements are stored in an array are in increasing order.
Input Format	First line contain the value of N. Second line contain the values of N element. Third line contain the element whose location to be find.
Output Format	Display the location.
Sample Input	6 10 12 15 18 20 25 18
Sample Output	4

6.3	Matrix Multiplication
Problem Statement	Write a program to multiply two matrices and display the resultant matrix. (Validate that number of columns of A = number of rows of B.)
Input Format	First line contain the order of two matrix. Second line contain the elements of first matrix. Third line contain the elements of second matrix.
Output Format	Display the resultant matrix.
Sample Input	<pre>2 3 3 3 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1</pre>
Sample Output	<pre>3 3 3 3 3 3</pre>